

Gradient Method

Shuo Guo

School of Science and Engineering
The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen), China
E-mail: shuoguo@link.cuhk.edu.cn

June 17, 2026

Contents

1	Introduction	2
1.1	Problem Setup	2
1.2	Gradient Method	2
2	L-smooth, convex and strongly convex	3
2.1	L-smooth	3
2.1.1	Learning Rate Choice	3
2.1.2	Convergence Rate	3
2.2	Convex+Smooth	4
2.2.1	Lower Bound	4
2.2.2	Upper Bound (Convergence Rate)	4
2.3	Strongly Convex	5
2.3.1	Lower Bound	5
2.3.2	Upper Bound	5
3	More general case	6
3.1	Coercivity	6
3.2	Gradient Linear Growth	6
3.3	Function-value Structure	6
3.4	One-step Descent	6
3.5	L	7
4	Ball	7
5	Proof	8

1. Introduction

I'm learning Optimization Methods, and want to have a note for Gradient Method (GM).

In [Chapter 1](#), I would introduce GM's definition. In [Chapter 2](#), I would discuss some simple case, including L -smooth, convex and strongly convex. And we will talk about coercivity and other more general case in [Chapter 3](#). (Actually, there is not too much gap between Chapter 2 and Chapter 3. Just because Chapter could be easier.) Also, in most cases, many condition cannot be satisfied globally, but locally, so we can pick up a ball, and analysis in the ball. I will show it in [Chapter 4](#). In these chapters, I would just provide some simple proof, and the whole proof will be in [Chapter 5](#). However, I'm not familiar with norm and other knowledge as I've not taken advanced algebra and analysis course, so I would not focus on that.

1.1. Problem Setup

Consider a function $f : \mathbb{R}^m \rightarrow \mathbb{R}$, and f is differentiable. We consider the first order oracle, which means:

$$\mathbb{R}^m \ni x \rightarrow \boxed{\text{First order oracle}} \rightarrow \nabla f(x) \in \mathbb{R}^m$$

Now we want to solve this problem:

$$\min_{x \in \mathbb{R}^m} f(x)$$

We define some notions: x^* means the solution¹, and $f^* = f(x^*)$.

1.2. Gradient Method

We would use GM to solve this problem:

Given $x_0 \in \mathbb{R}^m$ and a sequence $\{\gamma_n\}_{n \in \mathbb{N}} \subseteq [0, \infty)$. Thus, we have the rule:

$$x_n = x_{n-1} - \gamma_n \nabla f(x_{n-1})$$

However, one should notice that this problem is unsolvable if we don't give more information on the structure of f .

¹However, it could be local or global. Even worse, it could be saddle point in some 'bad' case.

2. L-smooth, convex and strongly convex

2.1. L-smooth

The definition of L -smooth:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$$

2.1.1 Learning Rate Choice

As f is L -smooth, we have:

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2.$$

And do further calculation, and let $f(x_{k+1}) \leq f(x_k) - \gamma_{k+1}\|\nabla f(x_k)\|^2$ we get:

$$f(x_{k+1}) \leq f(x_k) - \gamma_{k+1} \left(1 - \frac{L \cdot \gamma_{k+1}}{2}\right) \|\nabla f(x_k)\|^2$$

So if $0 < \gamma_{k+1} < \frac{2}{L}$, we can say $f(x_{k+1})$ must be smaller than $f(x_k)$.

We choose $\gamma = \frac{1}{L}$ for all $\{\gamma_n\}_{n \in \mathbb{N}}$. Thus, we have:

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L} \|\nabla f(x_k)\|^2$$

2.1.2 Convergence Rate

We could sum up the square of the derivatives, and use L -smooth to bound, and we get:

$$\min_{0 \leq k \leq N-1} \|\nabla f(x_k)\| \leq \sqrt{\frac{2L(f(x_0) - f^*)}{N}}$$

Another way to write is to calculate how many steps we at most need to get to the error less than ϵ :

$$N = \left\lceil \frac{2L(f(x_0) - f^*)}{\epsilon^2} \right\rceil$$

The whole proof is here: [PROOF](#)

2.2. Convex+Smooth

Definition is the following: L -smooth:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \quad \forall x, y.$$

Convex:

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle, \quad \forall x, y$$

2.2.1 Lower Bound

For all first order oracle method, in convex + smooth case, the convergence rate for f will not be larger than $O(\frac{1}{K^2})$

That means, for all first order oracle method, we can always create a very bad function, which makes the convergence rate cannot be larger than a certain rate. The function is:

$$f_p(x) = \frac{L}{4} \left\{ \frac{1}{2} \left[(x^{(1)})^2 + \sum_{i=1}^{p-1} (x^{(i)} - x^{(i+1)})^2 + (x^{(p)})^2 \right] - x^{(1)} \right\}, \quad x_0 = 0$$

The main idea of the function is that the oracle could know the gradient of the k -th elementary at least after the k -th step. The elementaries are coupled, and even we can achieve the best every step, the known information is still limited.

We could use the space of the gradient, and see how it compose x_k .

$$x_k \in \mathbb{R}^{k,n} := \{x \in \mathbb{R}^n \mid x^{(i)} = 0 \text{ for } i > k\}. \quad (11)$$

That is, after k iterations, the algorithm can only access the first k coordinates.

Now we analysis the convergence rate:

$$f(x_k) - f^* \geq \frac{3L\|x_0 - x^*\|^2}{32(k+1)^2} = O\left(\frac{1}{k^2}\right).$$

$$\|x_k - x^*\|^2 \geq \frac{1}{8}\|x_0 - x^*\|^2$$

Here is the proof for its lower bound: [PROOF](#).

2.2.2 Upper Bound (Convengence Rate)

2.3. Strongly Convex

2.3.1 Lower Bound

2.3.2 Upper Bound

3. More general case

Here we first give some structures without giving additional information. Then we give a function to measure the 'energy' of the 'system', to see how the structure lead to convergence. Last, we explain the connections of these structures.

3.1. Coercivity

The definition of Coercivity is: any x , we have:

$$\langle x - x^*, \nabla f(x) \rangle \geq c \|x - x^*\|^2$$

It means that the direction of the gradient at x will not be too 'bad'. It doesn't mean the gradient will point exactly to the optimal point. Just a rough description.

3.2. Gradient Linear Growth

The definition: for any x , we have:

$$\|\nabla f(x)\| \leq L \|x - x^*\|$$

Somehow it's like L -smooth, but it's a weaker description. We try to bound the gradient, so that it will not overshoot.

3.3. Function-value Structure

Consider a ray from x^* : $x^* + t(x - x^*)$, then we have:

$$f(x^* + t(x - x^*)) - f(x^* + s(x - x^*)) \geq (t^2 - s^2) \|x - x^*\|^2$$

3.4. One-step Descent

There exists $\epsilon \in [0, 1]$ s.t.

$$\|x_k - \gamma_{k+1} \nabla f(x) - x^*\|^2 \leq \epsilon \|x_k - x^*\|^2$$

3.5. L

4. Ball

5. Proof

Convergence for L-smooth

Theorem 5.1 (Global Convergence Rate). *Assume f is bounded below by $f^* > -\infty$. Let $\{x_k\}$ be generated by gradient descent with fixed step size $h = 1/L$. Then for any $N \geq 1$,*

$$\min_{0 \leq k \leq N-1} \|\nabla f(x_k)\| \leq \sqrt{\frac{2L(f(x_0) - f^*)}{N}}.$$

Consequently, the algorithm converges sublinearly at a rate of $O(1/\sqrt{N})$.

Proof. Summing the inequality from Theorem ?? for $k = 0, 1, \dots, N-1$, we obtain a telescoping series:

$$\sum_{k=0}^{N-1} \|\nabla f(x_k)\|^2 \leq 2L \sum_{k=0}^{N-1} (f(x_k) - f(x_{k+1})) = 2L(f(x_0) - f(x_N)).$$

Since $f(x_N) \geq f^*$, we have

$$\sum_{k=0}^{N-1} \|\nabla f(x_k)\|^2 \leq 2L(f(x_0) - f^*).$$

Now let $g_N^* := \min_{0 \leq k \leq N-1} \|\nabla f(x_k)\|$. Then

$$(g_N^*)^2 \leq \frac{1}{N} \sum_{k=0}^{N-1} \|\nabla f(x_k)\|^2 \leq \frac{2L(f(x_0) - f^*)}{N}.$$

Taking square roots yields the desired bound. □

Theorem 5.2 (Iteration Complexity to Reach an ϵ -Critical Point). *For any desired accuracy $\epsilon > 0$, gradient descent with $h = 1/L$ finds a point x_k such that $\|\nabla f(x_k)\| \leq \epsilon$ within at most*

$$N = \left\lceil \frac{2L(f(x_0) - f^*)}{\epsilon^2} \right\rceil$$

iterations.

Proof. To guarantee $g_N^* \leq \epsilon$, it suffices by Theorem 5.1 that

$$\sqrt{\frac{2L(f(x_0) - f^*)}{N}} \leq \epsilon.$$

Solving for N :

$$\frac{2L(f(x_0) - f^*)}{N} \leq \epsilon^2 \iff N \geq \frac{2L(f(x_0) - f^*)}{\epsilon^2}.$$

Thus the smallest integer N satisfying this bound is as stated. □

Proof for a extremely coupled function

We now show that the rate $O(1/N)$ cannot be improved to $o(1/N^2)$ in the worst case, and moreover, $\|x_k - x^*\|$ may never converge to 0.

We use the standard construction from Theorem 2.1.7 of Nesterov's lectures. Define the quadratic function for $p \geq 1$:

$$f_p(x) = \frac{L}{4} \left\{ \frac{1}{2} \left[(x^{(1)})^2 + \sum_{i=1}^{p-1} (x^{(i)} - x^{(i+1)})^2 + (x^{(p)})^2 \right] - x^{(1)} \right\}. \quad (8)$$

Step 1: Verify convexity and L-smoothness

The Hessian of f_p is $\nabla^2 f_p(x) = \frac{L}{4} A_p$, where A_p is the tridiagonal matrix with 2's on the diagonal and -1 's on the off-diagonal (except the first and last diagonal entries are 1). For any $h \in \mathbb{R}^n$,

$$\langle \nabla^2 f_p(x) h, h \rangle = \frac{L}{4} \left[(h^{(1)})^2 + \sum_{i=1}^{p-1} (h^{(i)} - h^{(i+1)})^2 + (h^{(p)})^2 \right].$$

Clearly this is ≥ 0 , so f_p is convex. Also,

$$\langle \nabla^2 f_p(x) h, h \rangle \leq L \sum_{i=1}^n (h^{(i)})^2 = L \|h\|^2,$$

so f_p is L -smooth. Thus $f_p \in \mathcal{F}_L^{\infty,1}(\mathbb{R}^n)$.

Step 2: Compute the unique minimizer and optimal value

Solving $\nabla f_p(x) = 0$ gives the unique minimizer:

$$\bar{x}_p^{(i)} = 1 - \frac{i}{p+1}, \quad i = 1, \dots, p, \quad \text{and} \quad \bar{x}_p^{(i)} = 0 \text{ for } i > p.$$

The optimal value is

$$f_p^* = -\frac{L}{8} \left(1 - \frac{1}{p+1} \right).$$

Also, from the norm bound,

$$\|\bar{x}_p\|^2 = \sum_{i=1}^p \left(1 - \frac{i}{p+1} \right)^2 \leq \frac{p+1}{3}.$$

Step 3: Information restriction (Key Lemma)

Let $x_0 = 0$. For any first-order method satisfying

$$x_k \in \text{Lin}\{\nabla f_p(x_0), \dots, \nabla f_p(x_{k-1})\},$$

we have by induction (due to the tridiagonal structure of A_p):

$$x_k \in \mathbb{R}^{k,n} := \{x \in \mathbb{R}^n \mid x^{(i)} = 0 \text{ for } i > k\}$$

That is, after k iterations, the algorithm can only access the first k coordinates.

Step 4: Lower bound for function values

Fix any $k \geq 1$ and choose $p = 2k + 1$. Apply the method to $f = f_{2k+1}$. By (11), $x_k \in \mathbb{R}^{k,n}$, so

$$f_{2k+1}(x_k) = f_k(x_k) \geq f_k^*.$$

Therefore,

$$f(x_k) - f^* = f_{2k+1}(x_k) - f_{2k+1}^* \geq f_k^* - f_{2k+1}^*.$$

Using (9):

$$\begin{aligned} f_k^* - f_{2k+1}^* &= -\frac{L}{8} \left(1 - \frac{1}{k+1} \right) + \frac{L}{8} \left(1 - \frac{1}{2k+2} \right) \\ &= \frac{L}{8} \left(\frac{1}{2k+2} - \frac{1}{k+1} \right) \\ &= \frac{L}{16(k+1)}. \end{aligned}$$

Since $x_0 = 0$ and $x^* = \bar{x}_{2k+1}$, from (10):

$$\|x_0 - x^*\|^2 = \|\bar{x}_{2k+1}\|^2 \leq \frac{2k+2}{3} = \frac{2(k+1)}{3}.$$

Hence

$$\frac{f(x_k) - f^*}{\|x_0 - x^*\|^2} \geq \frac{L}{16(k+1)} \cdot \frac{3}{2(k+1)} = \frac{3L}{32(k+1)^2}.$$

Thus we obtain the explicit lower bound:

$$\boxed{f(x_k) - f^* \geq \frac{3L\|x_0 - x^*\|^2}{32(k+1)^2} = \Omega\left(\frac{1}{k^2}\right).} \quad (\text{LB-Func})$$

Step 5: Lower bound for distance to the minimizer

Using the same construction with $p = 2k + 1$, since $x_k \in \mathbb{R}^{k,n}$, the coordinates $k+1, \dots, 2k+1$ of x_k are zero. Thus

$$\|x_k - x^*\|^2 \geq \sum_{i=k+1}^{2k+1} (\bar{x}_{2k+1}^{(i)})^2.$$

Substituting $\bar{x}_{2k+1}^{(i)} = 1 - \frac{i}{2k+2}$, we compute:

$$\begin{aligned} \sum_{i=k+1}^{2k+1} \left(1 - \frac{i}{2k+2} \right)^2 &= (k+1) - \frac{1}{k+1} \sum_{i=k+1}^{2k+1} i + \frac{1}{4(k+1)^2} \sum_{i=k+1}^{2k+1} i^2 \\ &= \frac{2k^2 + 7k + 6}{24(k+1)} \\ &\geq \frac{1}{8} \|\bar{x}_{2k+1}\|^2. \end{aligned}$$

The last inequality follows from $\|\bar{x}_{2k+1}\|^2 \leq \frac{2k+2}{3}$ and a direct comparison. Hence,

$$\boxed{\|x_k - x^*\|^2 \geq \frac{1}{8} \|x_0 - x^*\|^2 = \Omega(1).} \quad (\text{LB-Dist})$$